Pak-Austria FachhochschuleInstitute of Applied Sciencesand Technology, Haripur

Lab Task 10

Class:	BSE FA23 GREEN and Blue	Date:	16th April 2026
Subject:	Software Design And Architecture Lab	Instructor:	Dr. Nabeel Ahmed
Time Allowed:	2 hr 30 Mins	Lab Engineer:	Zainab Tahir

Instructions:

You are required to complete the following lab task individually. **Plagiarism or copying from other students isstrictly prohibited.** Ensure that your work is original and reflects your understanding of the task.

You are developing a Cricket Team Management System using Spring Boot and MySQL. The system should store information about players in a cricket team such as playerId, name, role (batsman/bowler/all-rounder), and teamName.

Task:

Using **Spring Data JPA**, design and implement the following:

1. Create a **Player Entity class** and map it to a database table.
2. Create a **Repository interface** by extending JpaRepository.

3. Configure the application to connect with a **MySQL database**.
4. Implement methods to:
    - Add a new player
    - Get all players
    - Find players by team name
    - Delete a player by ID
5. Explain how JPA automatically performs CRUD operations without writing SQL queries